

Example of a non-full rank s-parameter exponential family distribution.

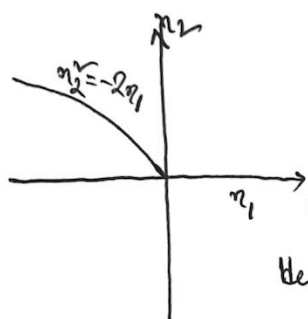
r.v. $X \sim N(\theta, \theta^2)$, $\theta > 0$, $\Theta = \{\theta: \theta > 0\}$.

$$f_{\theta}(x) = \frac{1}{\theta\sqrt{2\pi}} \exp. -\frac{1}{2\theta^2}(x-\theta)^2$$

$$= \frac{e^{-1/2}}{\sqrt{2\pi}} \exp. \left(-\frac{1}{2\theta^2}x^2 + \frac{1}{\theta}x - \ln \theta\right), \text{ so that}$$

$$\eta_1(\theta) = \eta_1 = -\frac{1}{2\theta^2}, \quad T_1(x) = x^2$$

$$\eta_2(\theta) = \eta_2 = \frac{1}{\theta}, \quad T_2(x) = x$$



The natural parameter space of the minimal canonical form is $H = \{(\eta_1, \eta_2) : \eta_2^2 = -2\eta_1, \eta_1 < 0, \eta_2 > 0\}$, which is a curve and does not contain an open rectangle of dim. 2

Hence not a full-rank exponential family distn

Imp Note the result (R) is a sufficient (or 'if') condition statement and not a necessary (or 'only if') one.

Hence, for a r.v. $X_1, \dots, X_n$ iid from $N(\theta, \theta^2)$, $\theta > 0$, to show that the MSS $T(X) = (\sum X_i, \sum X_i^2)$ is not complete, we have to go by the Defn of completeness of a statistic. (to show $\exists$ a fn of the MSS T , $\exists E_0(h(T)) = 0 \forall \theta \in \Theta \nRightarrow h(t) = 0$ a.e.)

In this case, $\sum_{i=1}^n X_i \sim N(n\theta, n\theta^2)$, $T(X) = (\sum X_i, \sum X_i^2)$ is MS.

$$\Rightarrow E[(\sum X_i)^2] = V(\sum X_i) + [E(\sum X_i)]^2 = n\theta^2 + n^2\theta^2 = n(n+1)\theta^2 \neq 0$$

$$\Rightarrow E\left[\frac{(\sum X_i)^2}{n(n+1)}\right] = \theta^2 \neq 0$$

$$\text{Further, } E(X_i^2) = V(X_i) + E^2(X_i) = \theta^2 + \theta^2 = 2\theta^2$$

$$\sum E(X_i^2) = \sum 2\theta^2 = 2n\theta^2$$

$$\Rightarrow E(\sum X_i^2) = 2n\theta^2$$

$$\Rightarrow E\left(\frac{\sum X_i^2}{2n}\right) = \theta^2 \neq 0$$

so that $E\left(\frac{(\sum X_i)^2}{n(n+1)} - \frac{\sum X_i^2}{2n}\right) = 0 \forall \theta$, but $\frac{(\sum X_i)^2}{n(n+1)} - \frac{\sum X_i^2}{2n} \neq 0$ a.e.

in fact, $\frac{(\sum X_i)^2}{n(n+1)} \neq \frac{\sum X_i^2}{2n}$ w.p. 1

Exo Completeness of a MSS for a distribution not belonging to the exponential family of distns

$X_1, \dots, X_n$ iid $U(0, \theta)$, $\theta > 0$

$T = X_{(n)}$ is MS, and complete also (already shown).

The Lehman-Scheffe Theorem

Let T be a CSS & $\exists$ a unbiased estimator δ of $g(\theta)$.

Then $\exists$ a unique UMVUE of $g(\theta)$ given by $E(\delta(X)|T)$.

Proof Let δ_1 and δ_2 be unbiased estimators of $g(\theta)$

$\therefore E[\delta_1(X)|T]$ and $E[\delta_2(X)|T]$ are both unb. est. of $g(\theta)$,
(upto this 'sufficiency' of T is enough)
denoted by, say, $\eta_1(T)$ and $\eta_2(T)$.

$\therefore E_\theta[\eta_1(T) - \eta_2(T)] = 0 \quad \forall \theta \Rightarrow \eta_1(t) = \eta_2(t) \text{ a.e.}$ (this is possible because T is 'complete').

Hence, we get uniqueness of the estimator

Further, by the Rao-Blackwell this estimator has the min var. among all u.e.s of $g(\theta)$.

Hence, $E[\delta(X)|T]$ is the unique UMVUE of $g(\theta)$.

Corollary of the theorem: Let X be distd according to a distn from $\mathcal{P} = \{f_\theta: \theta \in \Theta\}$.
 T is a CSS. Then if $g(\theta)$ is unbiasedly estimable, it has one and only one unbiased estimator that is a function of T , and this is the unique UMVUE of $g(\theta)$.

We discuss 3 distinct methods of obtaining the UMVUE of $g(\theta)$

Method I: By inspection/Guess Method

For a relatively simple $g(\theta)$, start with $E(T)$, T is CSS. Eventually $g(\theta)$ will be $E[h(T)]$, for a simple function $h(\cdot)$ of T

Ex 1 $X_1, \dots, X_n$ iid $\text{Ber}(\theta)$, $\theta \in (0,1)$, $g(\theta) = \theta$.
 $T = \sum_{i=1}^n X_i$ is MSS. is complete (' $\text{Ber}(\theta)$ ' is 1-pm exp. family with full rank)
Further, $E(\frac{\sum X_i}{n}) = \theta$. $\therefore \frac{\sum X_i}{n} = \bar{X}$ is the unique UMVUE of $g(\theta) = \theta$.

Ex 2 $X_1, \dots, X_n$ iid $N(0,1)$, $\theta \in \mathbb{R}$, $g(\theta) = \theta$.

$T = \sum X_i$ MSS is complete (1-pm exponential family with full rank)

Further, $E_\theta(T) = \sum E(X_i) = n\theta \Rightarrow \frac{\sum X_i}{n} = \bar{X}$ is the unique UMVUE of $g(\theta) = \theta$.

Ex 3 $X_1, \dots, X_n$ iid $U(0,\theta)$, $\theta > 0$, $g(\theta) = \theta$

$T = X_{(n)}$ CSS (completeness from defn).

$E_\theta(T) = \frac{n}{\theta^{n+1}} \int_0^\theta t^n dt = \frac{n}{n+1} \theta \Rightarrow \frac{n+1}{n} X_{(n)}$ is the unique UMVUE of $g(\theta) = \theta$.

Method II. By solving / Direct Method

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T is CSS. Solve for a $\delta(T) \ni E_{\theta}(\delta(T)) = g(\theta) \forall \theta$

By Lehman Scheffe $\delta(T)$ is the unique UMVUE.
Theorem,

Ex $x_1, \dots, x_n$ iid $P(\theta)$, $\theta > 0$; $g(\theta) = e^{-\theta}$

$T = \sum x_i$ CSS. , mth $T \sim P(n\theta)$

We have $\sum_{t=0}^{\infty} \delta(t) \cdot e^{-n\theta} \frac{(n\theta)^t}{t!} = e^{-\theta} \forall \theta$

$$\Rightarrow \sum_{t=0}^{\infty} \delta(t) \frac{(n\theta)^t}{t!} = e^{(n-1)\theta} = \sum_{t=0}^{\infty} \frac{\{(n-1)\theta\}^t}{t!} \forall \theta$$

$$\Rightarrow \delta(t) \frac{n^t}{t!} = \frac{(n-1)^t}{t!} \text{ for } t=0, 1, \dots; n \geq 2$$

$$\Rightarrow \delta(T) = \left(\frac{n-1}{n}\right)^T \text{ is the unique UMVUE.}$$

Method III By conditioning / Rao-Blackwellization Method.

Ex, Find any unb. est. of $g(\theta)$, say, $\delta_0(x)$, then

$\delta(T) = E[\delta_0(x) | T]$ where T is CS, is the unique UMVUE of $g(\theta)$

Ex $x_1, \dots, x_n$ iid $P(\theta)$, $\theta > 0$; $g(\theta) = e^{-\theta}$

$T = \sum x_i \sim P(n\theta)$ is CS.

$\delta_0(x) = \begin{cases} 1 & \text{if } x_1 = 0 \\ 0 & \text{otherwise} \end{cases}$ then $\delta_0(x)$ is a unb. est. of $g(\theta)$.

$$\begin{aligned} \text{Then, } \delta(t) &= E[\delta_0(x) | T=t] = P(x_1=0 | T=t) = \frac{P(x_1=0, T=t)}{P(T)} = \frac{P(x_1=0, \sum_{i=1}^n x_i=t)}{P(\sum_{i=1}^n x_i=t)} \\ &= \frac{e^{-\theta} \cdot e^{-(n-1)\theta} \frac{(n-1)^{t-1}}{(t-1)!}}{e^{-n\theta} \frac{(n\theta)^t}{t!}} = \left(\frac{n-1}{n}\right)^t \end{aligned}$$

$$\Rightarrow \delta(T) = \left(\frac{n-1}{n}\right)^T, n \geq 2 \text{ is the unique UMVUE of } g(\theta) = e^{-\theta}.$$

A useful result for Method 3 is the Basu's Theorem, which states
Basu's Theorem For a family of distn $\mathcal{P} = \{P_{\theta}, \theta \in \Theta\}$, if T is a CSS
and V is an ancillary statistic, then V & T are independently
distributed.

Note the defn The statistic $V(x)$ is said to be ancillary if the distn of $V(x)$
does not depend on θ

ii) $V(x)$ is said to be first order ancillary if $E_{\theta}[V(x)]$ is
independent of θ .

Wlog, for an ancillary $V(x)$, $E_{\theta}(V(x)) = 0 \forall \theta \in \Theta$.